Lab 06 Manual

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SRT411NAA: Digital Data Analytics

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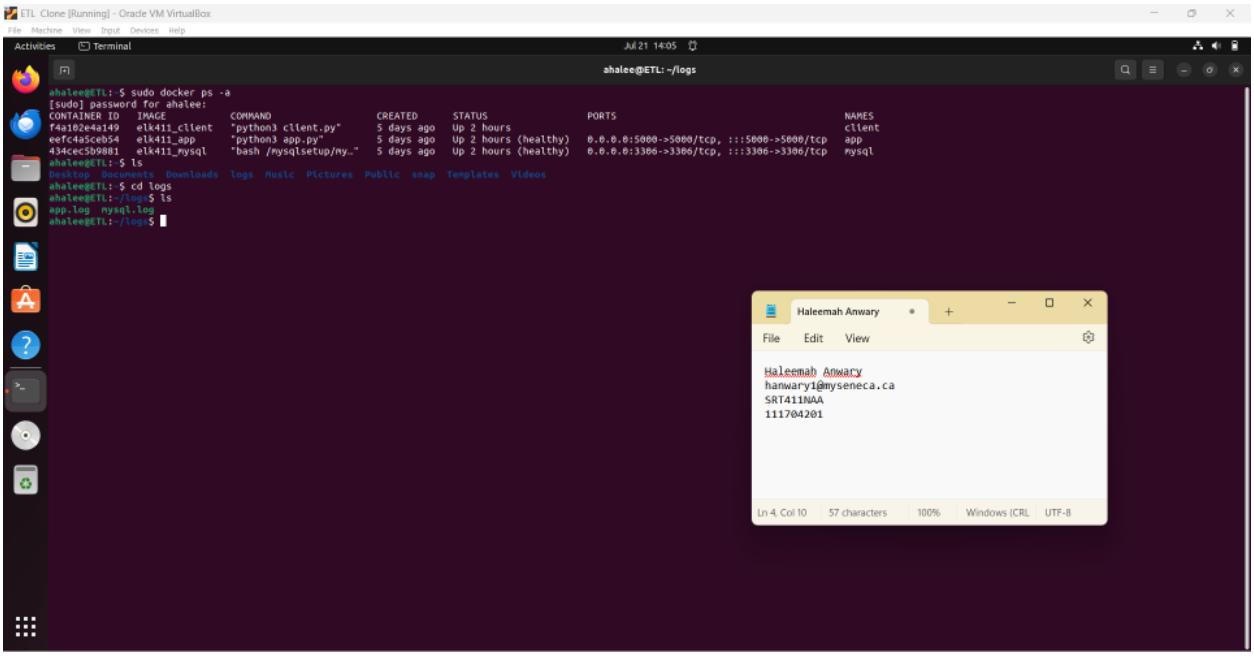
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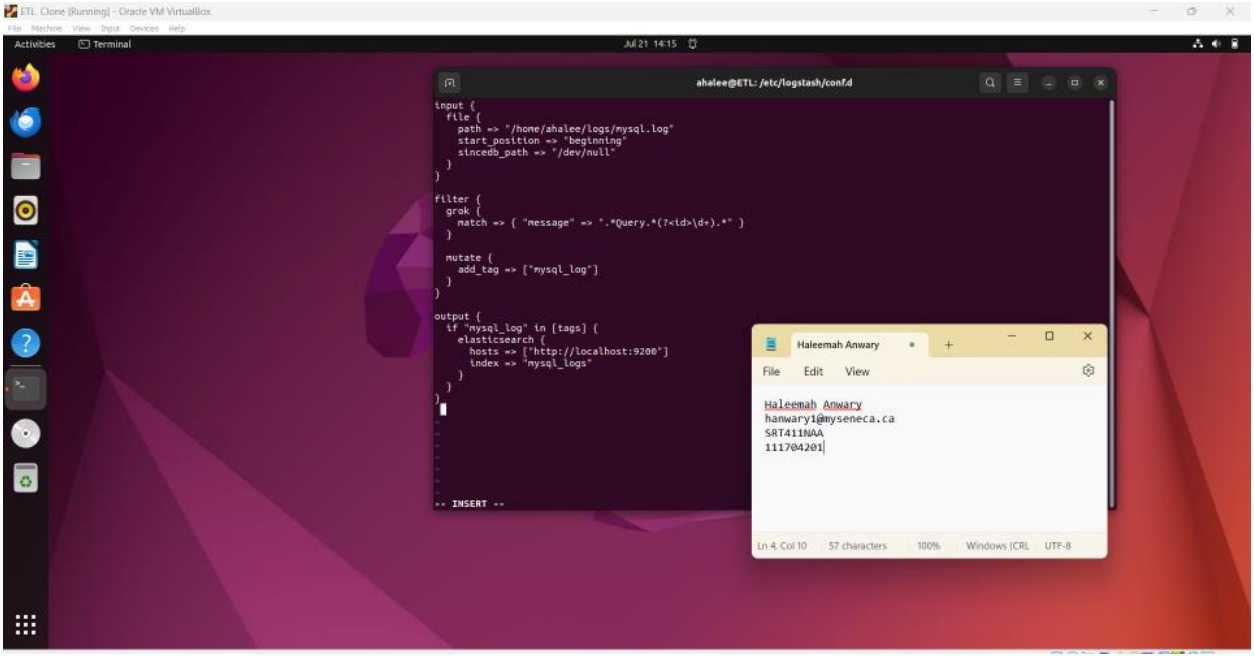
Centralised Logging (Task 01)

Run App Stack and extract logs

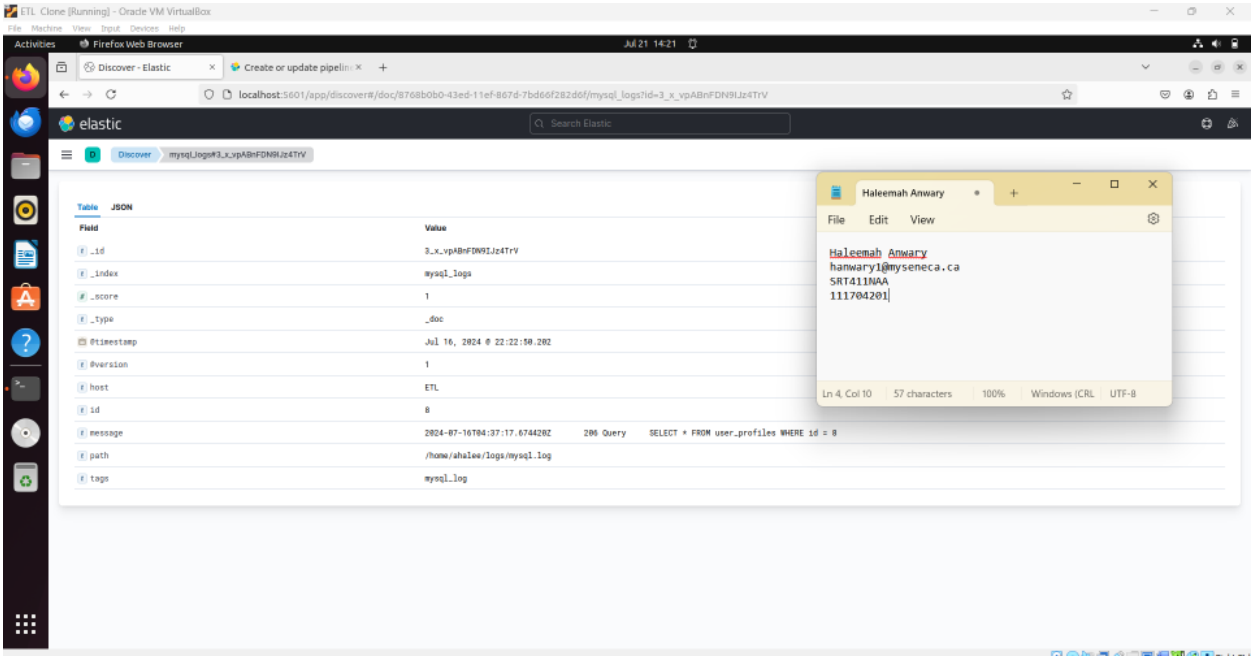


MYSQL Logstash pipeline

1. **Input:**
    - Logstash reads from the file /home/ahalee/logs/mysql.log, starting from the beginning each time.
2. **Filter:**
    - The grok filter extracts an id from lines containing "Query".
    - The mutate filter adds a mysql_log tag to these events.
3. **Output:**
    - Events with the mysql_log tag are sent to Elasticsearch, to the mysql_logs index on localhost:9200.



Data in Discover, we can see the extracted ‘id’ and the tag we initialized in the pipeline



MYAPP Logstash Pipeline

Explanation

Input Section:

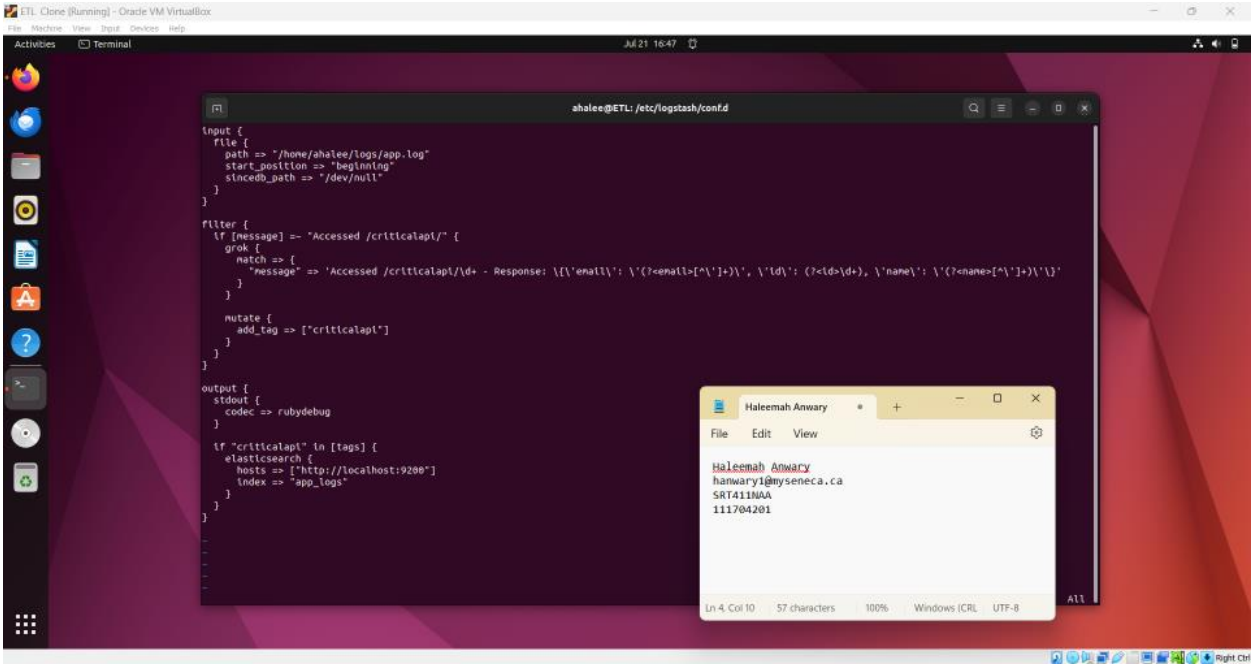
- Reads the log file from /var/log/myapp/app.txt starting from the beginning each time (start_position => "beginning").
- sincedb_path => "/dev/null" disables sincedb tracking.

Filter Section:

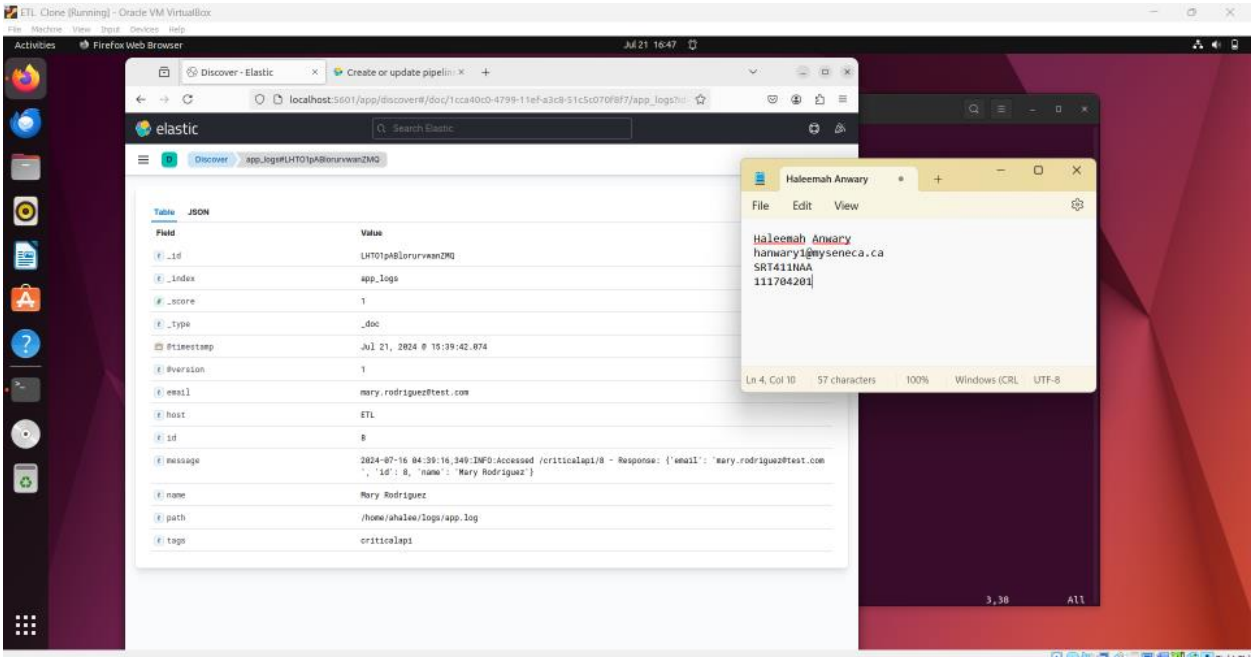
- Conditional Check: Checks if the message contains "Accessed /criticalapi/".
- Grok Filter: Matches the log structure:
    - Accessed /criticalapi/\d+ - Response: matches the initial part of the log message.
    - \'email\': \'(?<email>[^\']+)\' captures the email field.
    - \'id\': (?<id>\d+) captures the id field.
    - \'name\': \'(?<name>[^\']+)\' captures the name field.
- Mutate Filter: Adds the criticalapi tag to matched events.

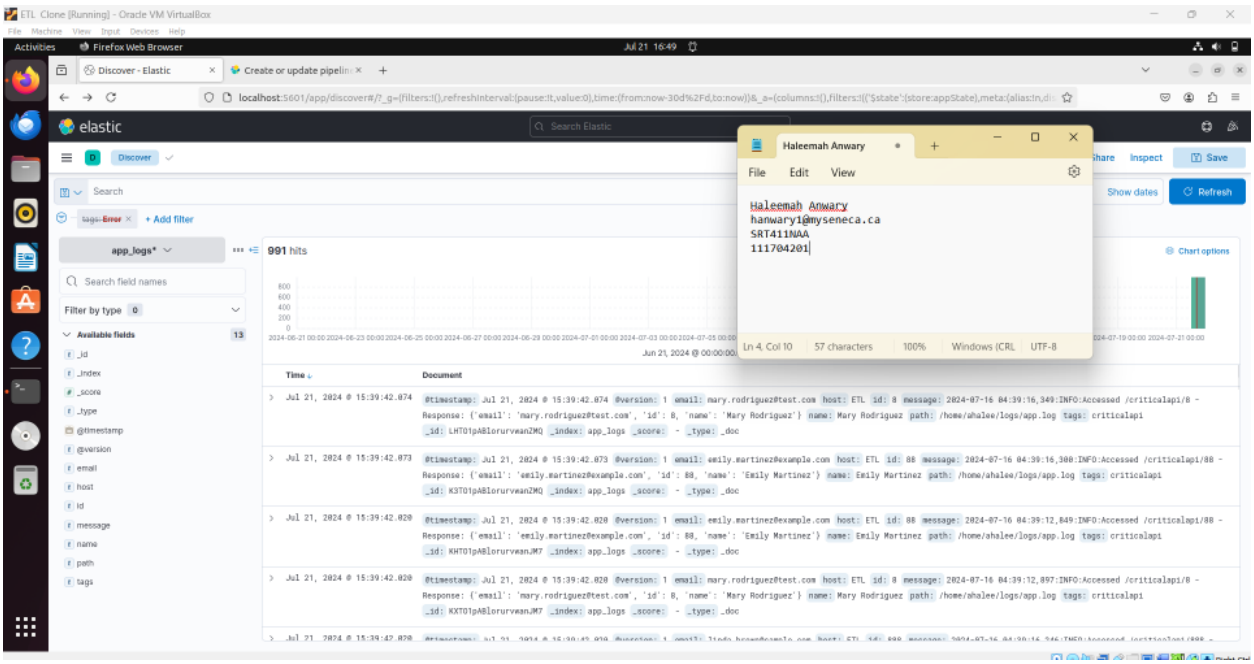
Output Section:

- stdout Output: Prints parsed events to the console with the rubydebug codec for debugging.
- Elasticsearch Output: Sends events with the criticalapi tag to Elasticsearch, indexing them in the app_logs index.

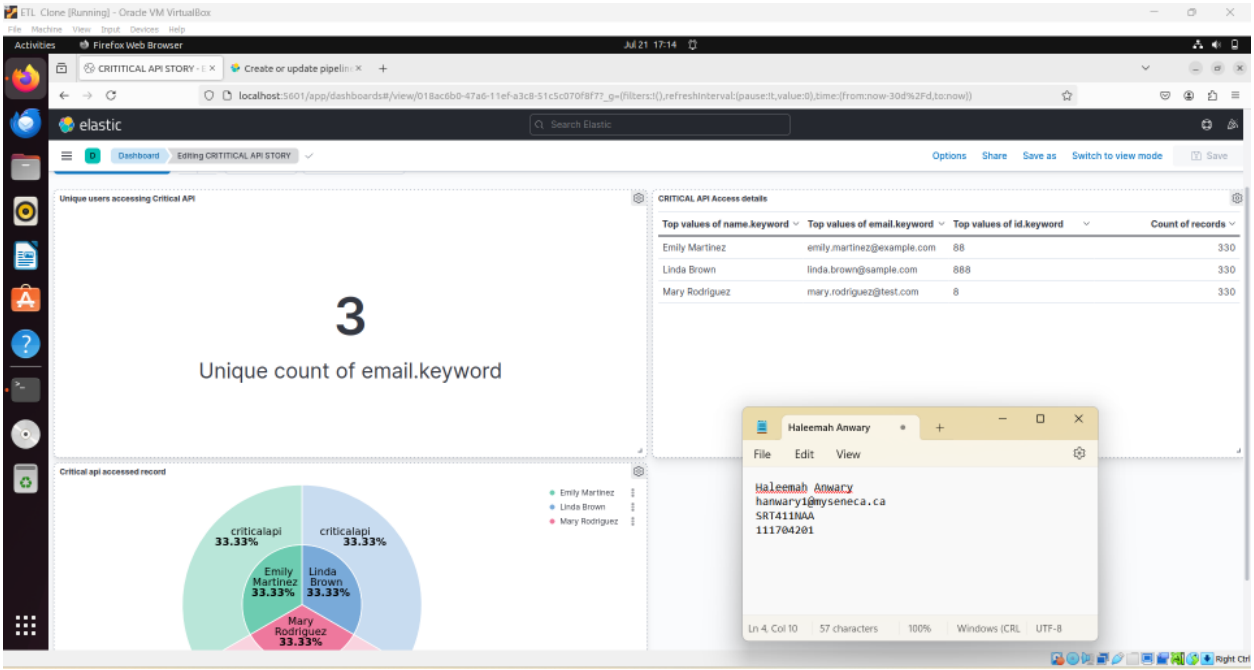


Data in Discover, we can see the extracted ‘id’, ‘name’, and ‘email’ as separate attributes and the tag.





Create single dashboards from the ingested data demonstration a data story of “criticalapi” exposure.



Centralised Logging Questions (Task 02)

Qualities of a Good Log Message

- **Clarity and Conciseness:** Clear and straightforward, no ambiguity.
- **Timestamp:** Precise timestamp for tracking events.
- **Contextual Information:** Includes source, user/session ID, request ID, and environment details.
- **Severity Level:** Uses appropriate levels (DEBUG, INFO, WARN, ERROR, CRITICAL) to indicate importance.
- **Actionability:** Provides enough detail to take action.
- **Structured Data**: Uses a structured format like JSON for easy parsing and analysis.
- **Readability:** Readable by both humans and machines.
- **Security:** Avoids sensitive information, complies with privacy regulations.
- **Error Details**: Includes error codes, messages, and context for quick issue resolution.
- **Consistency:** Maintains a consistent format and language

Descriptive Messages

Descriptive Message: A message that provides detailed information about an event, making it easy to understand what happened.

Good Mental Exercise: Imagine explaining the event to a colleague with no prior knowledge. Ensure your message covers the who, what, when, where, and why.

Relatable Logs

Relatable Log: A log that is easily understood and meaningful to its audience.

Headache with Servers: Different servers may have varying formats and contexts, making it hard to correlate events.

Increase Relatability: Use consistent formats, include contextual information, and standardize the logging practices across servers.

Complete Log Messages

Complete Log Message: Provides all necessary details, including the event description, context, and actionable information.

Differences:

- **Complete**: Includes who, what, when, where, why, and how.
- **Incomplete:** Lacks critical details, making it hard to understand or act upon.

Converting Log Files

Five rules for converting log files to friendlier formats:

Consistency: Ensure uniform formatting.

Structure: Use structured formats like JSON or CSV.

Clarity: Make messages clear and concise.

Context: Include contextual information.

Readability: Ensure logs are human-readable.

Categories of Log Files

1. **Application Logs**: Track application activities and errors.
2. **System Logs**: Record operating system and system service events.
3. **Security Logs**: Capture security-related events, such as login attempts and access control violations.

Security Logs in Systems

Logs to look at:

Authentication Logs: Monitor login attempts and access control.

Audit Logs: Track changes and actions performed on the system.

Application Security Logs: Capture security-related events within applications.

System Configuration, Events, and Logs

System Configuration: Settings and parameters that define how a system operates.

Events Generated: Activities or occurrences that are logged, such as user logins, system errors, and application activities.

Logs Collected: Records of the events generated, stored for monitoring and analysis.

Questions and Collected Logs

1. **What are the most frequently accessed API endpoints?**
    - **Logs Analysis:** By analyzing the application logs, we can identify which API endpoints are accessed most frequently, based on the URL patterns present in the log entries.
2. **Are there any unauthorized access attempts to critical endpoints?**

- **Logs Analysis:** The logs contain information about the status codes returned by the server. Unauthorized access attempts can be identified by looking for 4xx status codes, especially 401 (Unauthorized) and 403 (Forbidden).

3. **What are the response times for different API endpoints?**

    - **Logs Analysis**: If the logs contain timing information for requests and responses, this data can be used to calculate response times for each endpoint.

4. **Who are the users accessing the critical API endpoints?**

    - **Logs Analysis:** By examining the response details for critical API accesses, user information such as email and ID can be extracted to identify who is accessing these endpoints.

Describing Collected Logs

Application Logs

1. **Fields of Interest for Security Investigation**:
    - **Timestamp**: Indicates when the request was made.
    - **IP Address**: The source of the request.
    - **HTTP Method and URL**: Identifies the endpoint accessed.
    - **Response Status Code**: Indicates the success or failure of the request.
    - **Response Body**: Can contain sensitive user information and is critical for understanding the context of the request.
    - **Log Level (INFO)**: Indicates the type of log entry, which can help in filtering specific log levels during analysis.
2. **Log File Format**:
    - **Type of Data**: The logs are in plain text format, with each line containing:
        - Timestamp
        - Log level
        - IP Address
        - HTTP Method and URL
        - Response Status Code
        - Response Body (for certain endpoints)

MySQL Server Logs

1. **Fields of Interest for Security Investigation**:
    - **Timestamp**: Indicates when each action was performed.
    - **User and Host**: Identifies the user performing the action and the host from which the connection originated.
    - **Query**: The SQL statement executed, which is critical for detecting unauthorized access or malicious activities.

    - **Connection Information**: Details about the connections established and terminated.
2. **Log File Format**:
    - **Type of Data**: The logs are in plain text format, with each line containing:
        - Timestamp
        - Connection ID
        - Command (e.g., Connect, Quit, Query)
        - User and Host Information
        - SQL Query (if applicable)

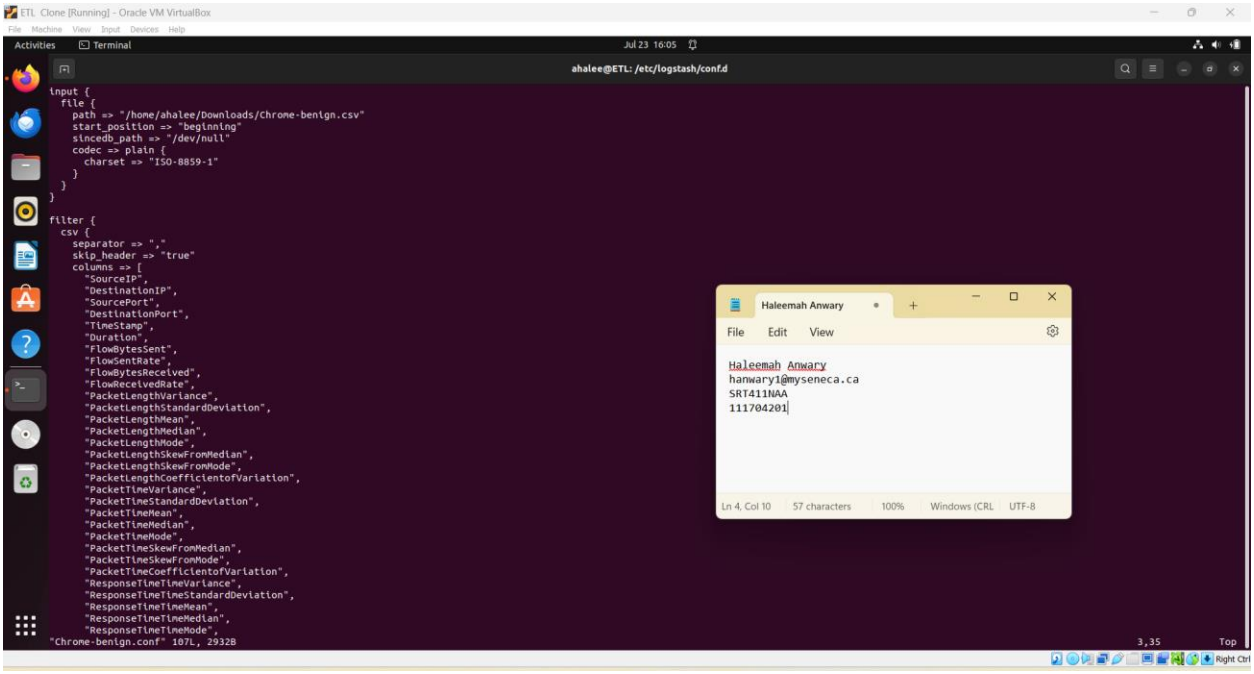
List of Events Generated and How Collected Logs Help Identify Them

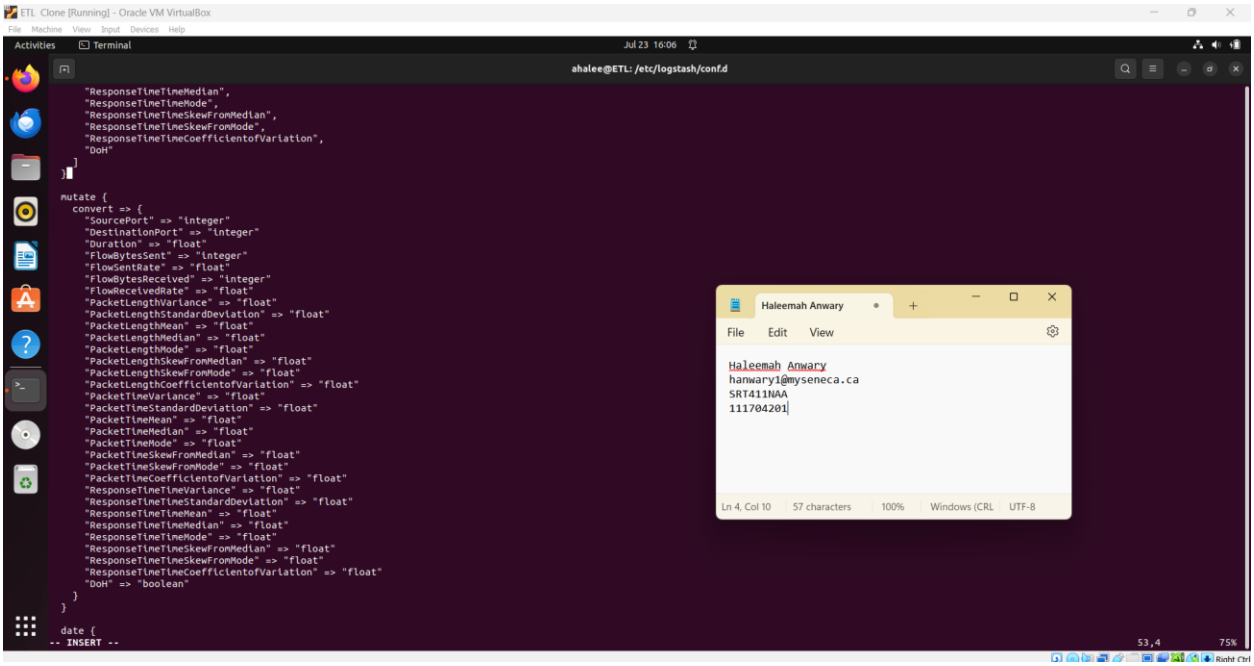
1. **Successful API Access**:
    - **Identification**: HTTP status code 200 in application logs.
2. **Failed API Access**:
    - **Identification**: HTTP status codes 4xx and 5xx in application logs.
3. **Database Connection Established**:
    - **Identification**: "Connect" entries in MySQL logs.
4. **Database Query Executed**:
    - **Identification**: "Query" entries in MySQL logs.
5. **User Creation or Privilege Changes**:
    - **Identification**: Specific SQL commands like CREATE USER, GRANT, etc., in MySQL logs.
6. **Data Insertion or Modification**:
    - **Identification**: INSERT, UPDATE, or DELETE commands in MySQL logs.

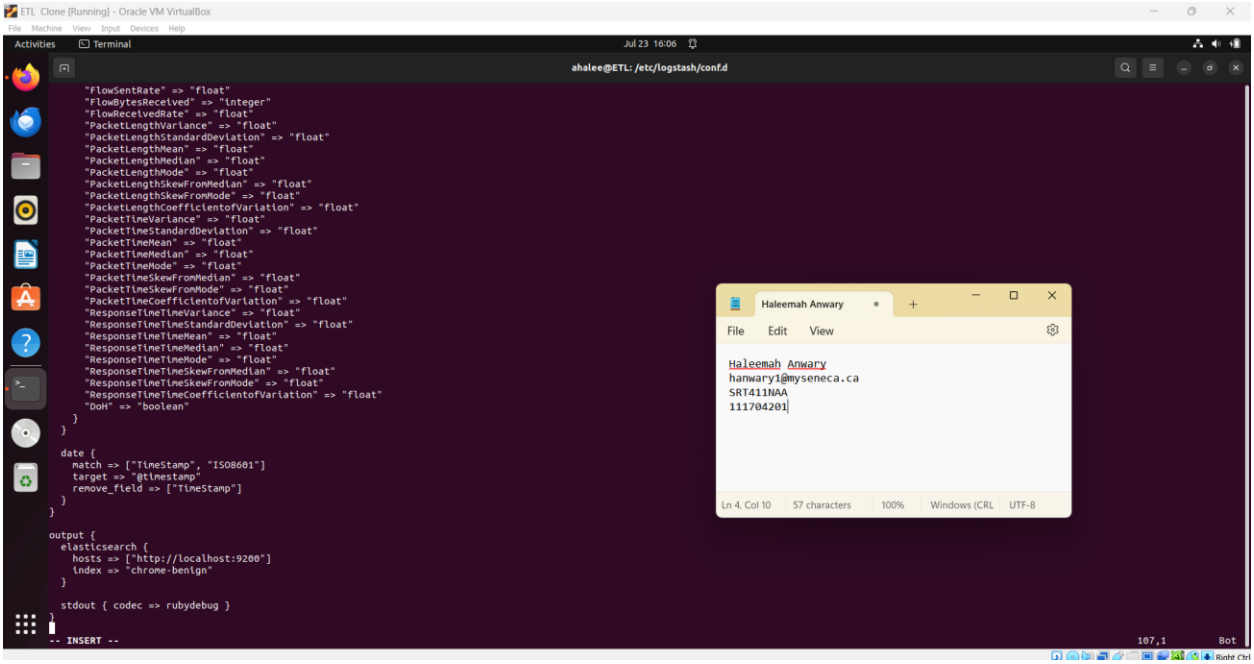
Data Ingestion, Statistical Analysis, and Visualization (Task 03)

Chrome-benign

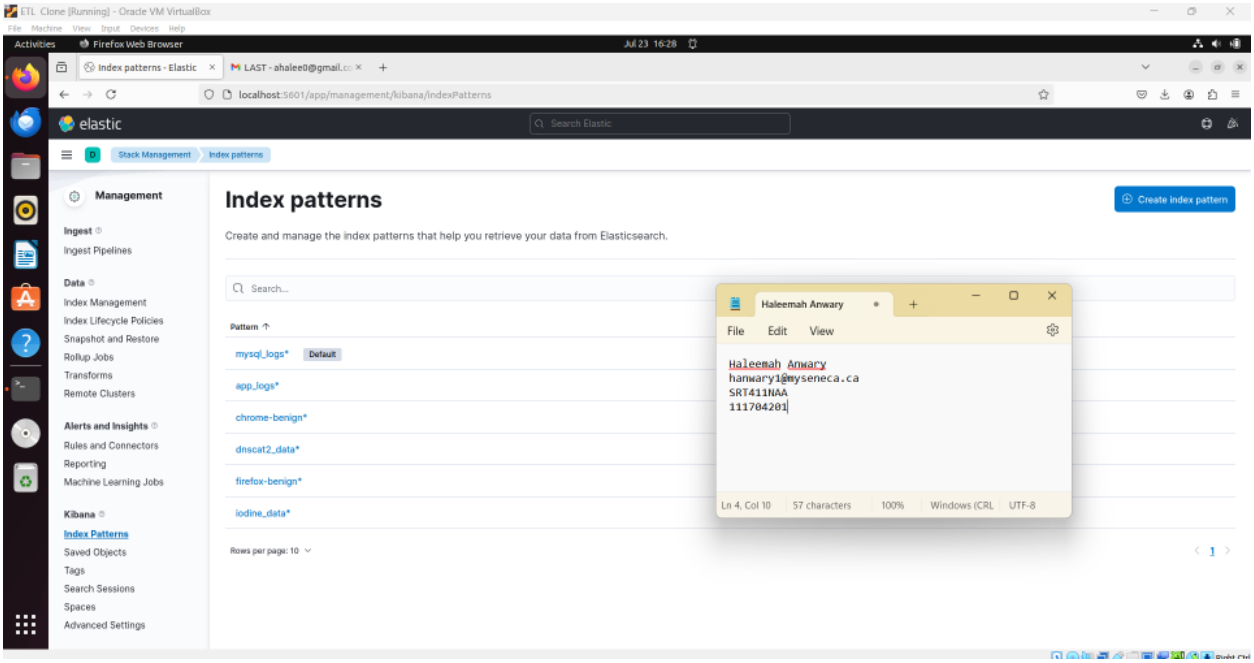
Logstash pipeline



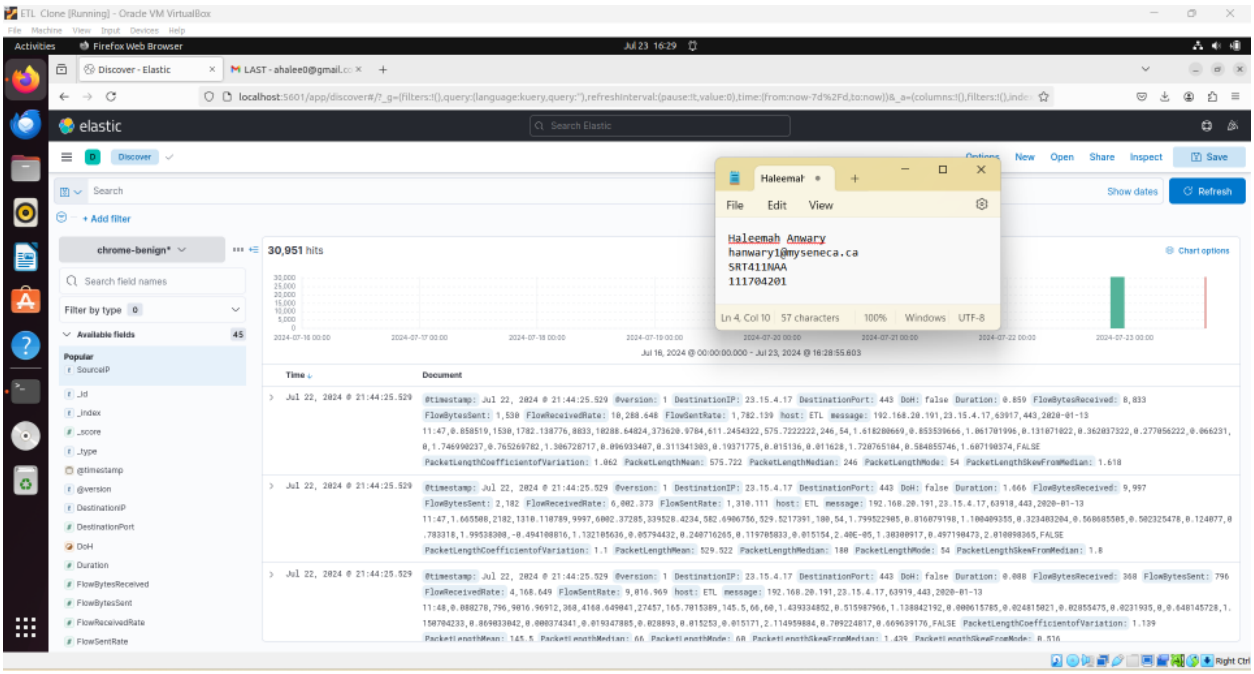




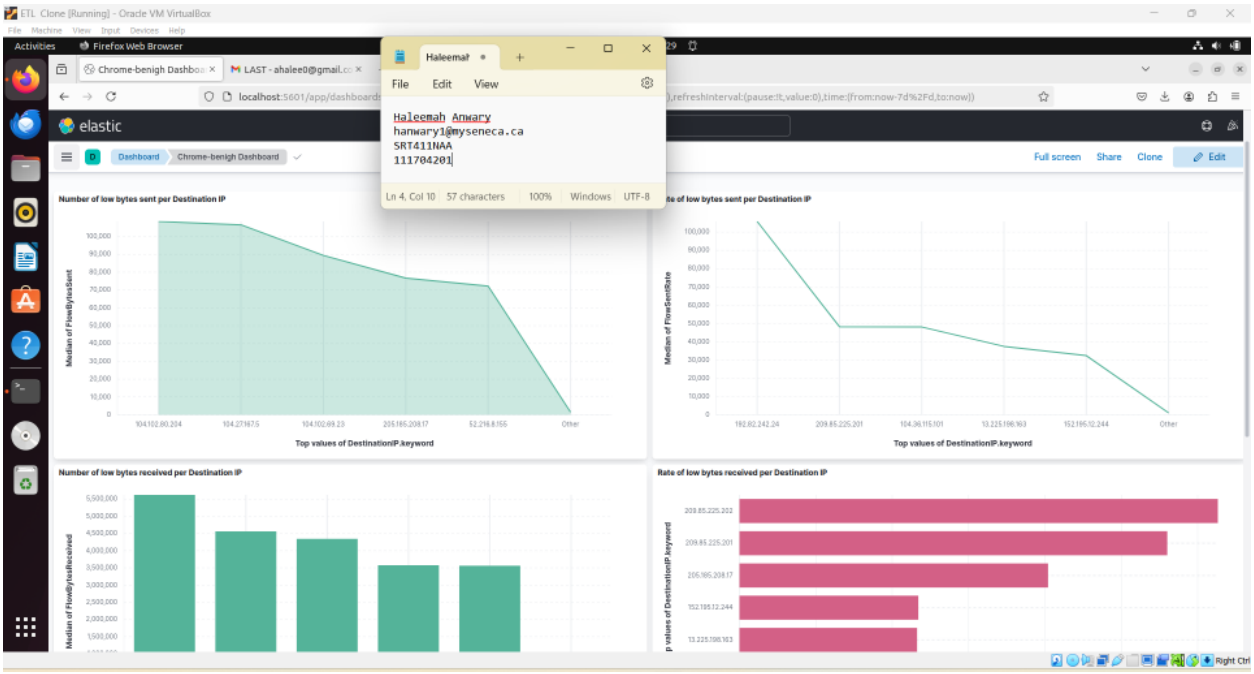
Index Patterns

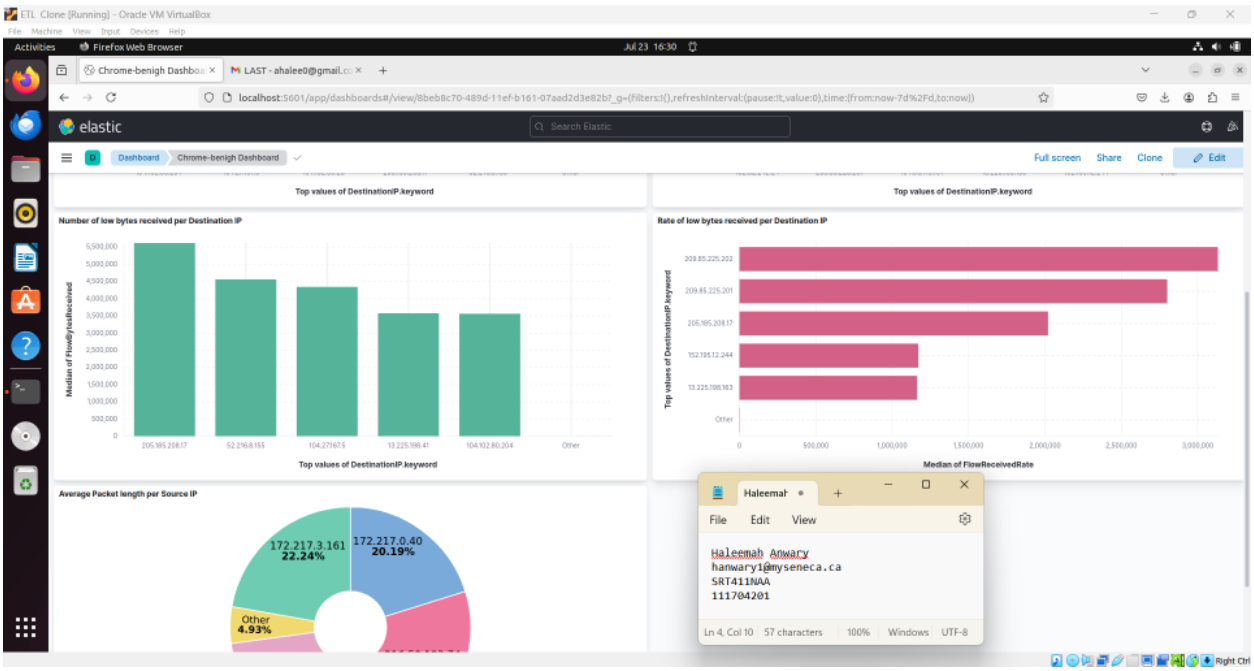


Data in Discover



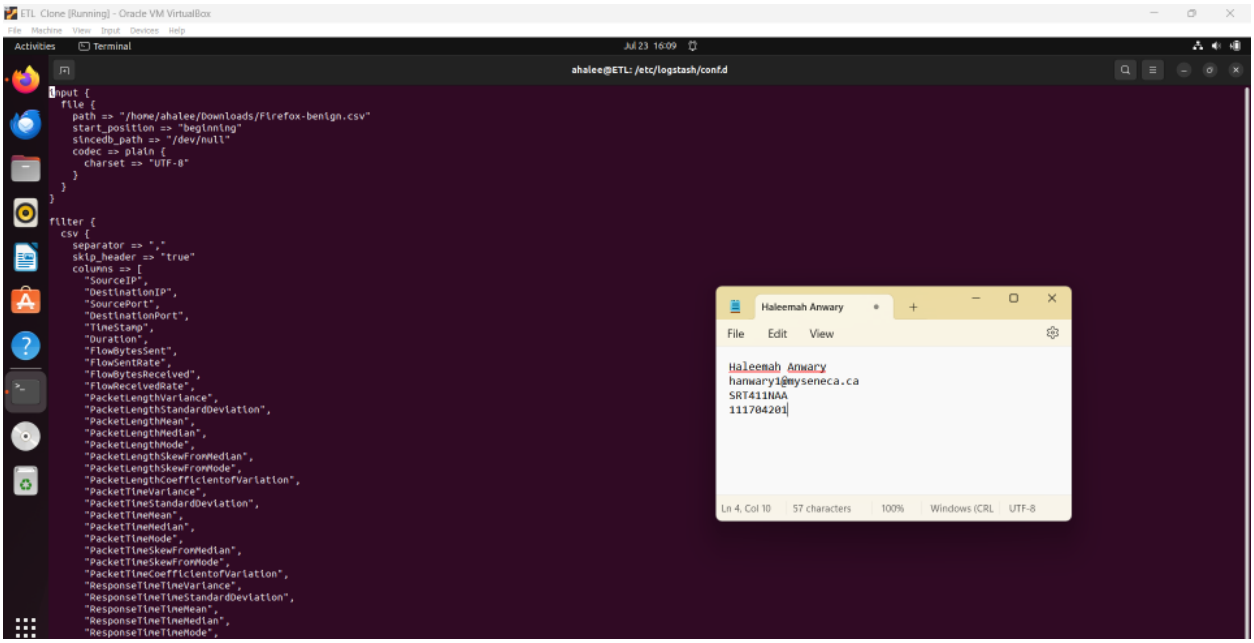
Visualizations

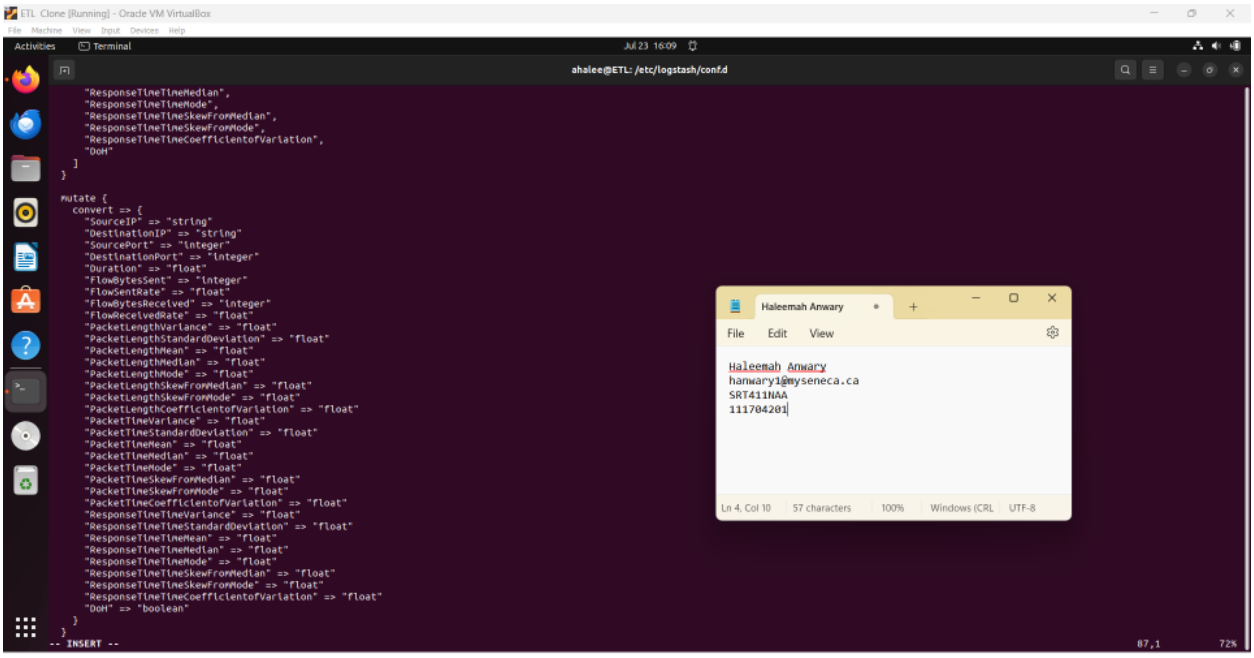


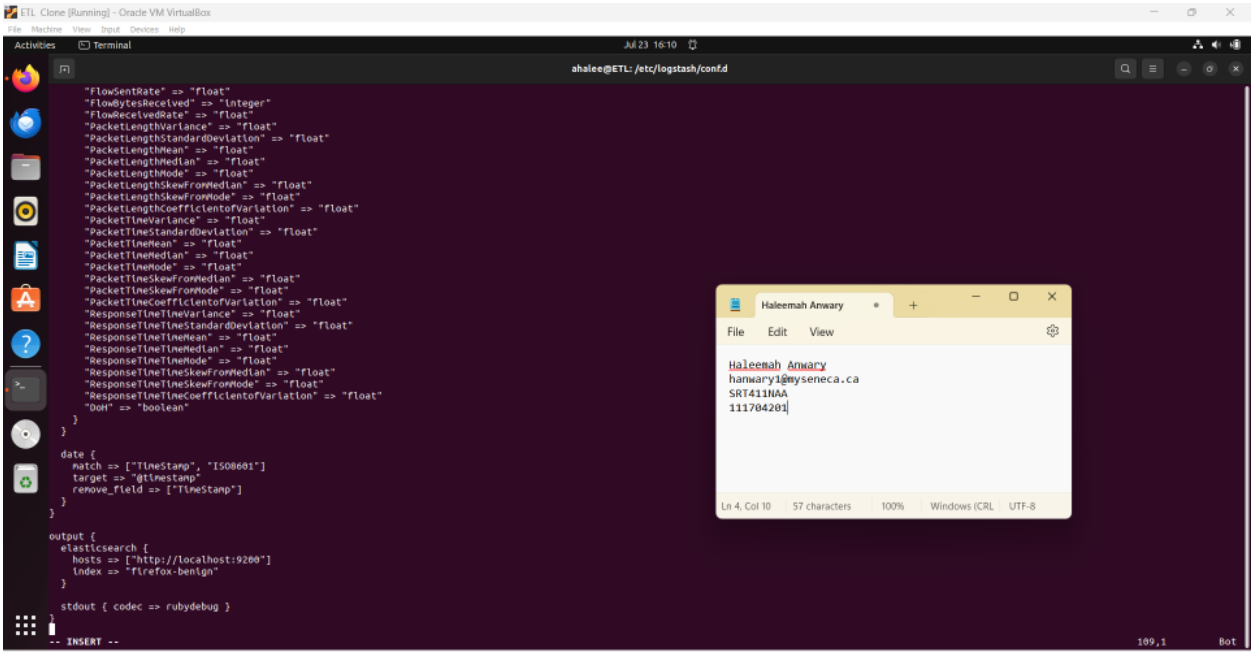


Firefox-benign

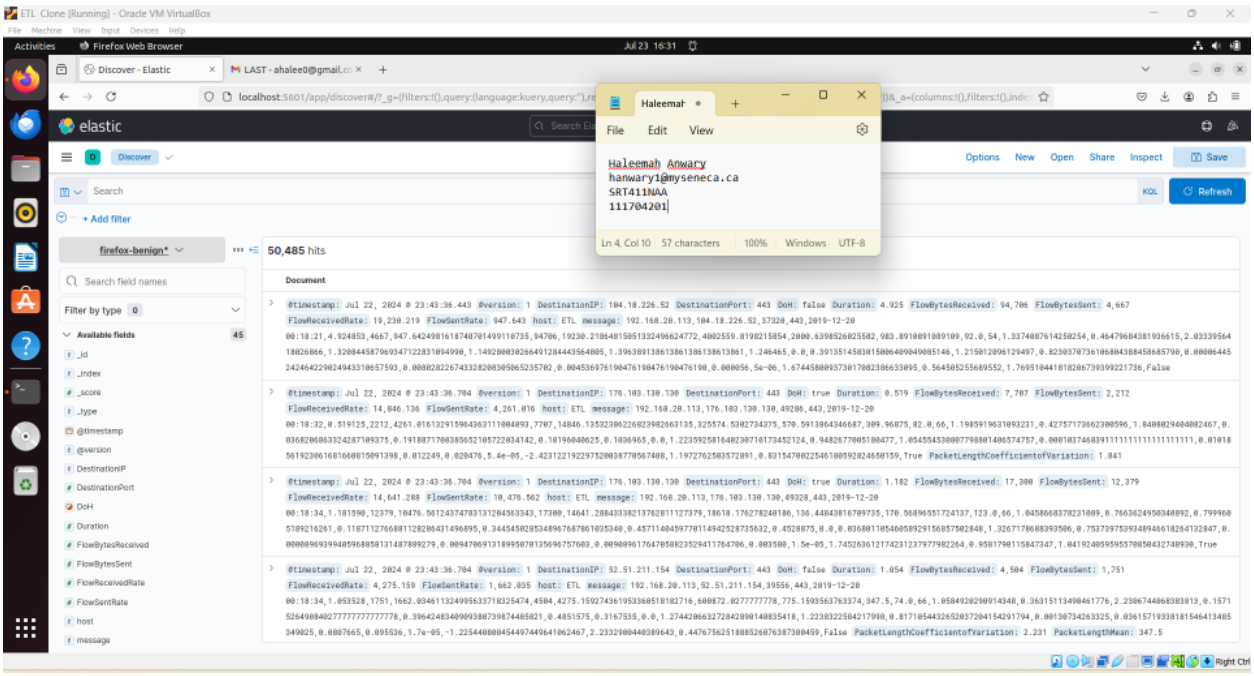
Logstash Pipeline



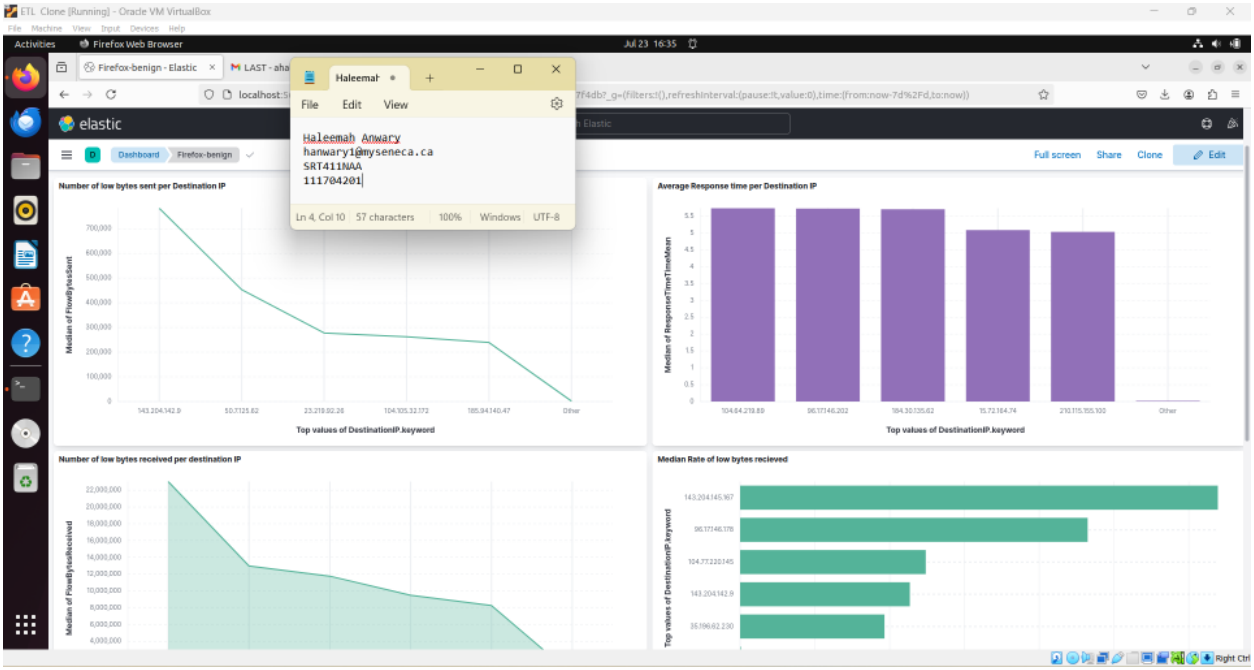

ETL Clone [Running] - Oracle VM VirtualBox
Activities
Terminal
Jul 23 16:09
ahalee@ETL: /etc/logstash/conf.d
"ResponseTimeTimeMedian",
"ResponseTimeTimeMode",
"ResponseTimeTimeSkewFromMedian",
"ResponseTimeTimeSkewFromMode",
"ResponseTimeTimeCoefficientofVariation",
"DoH"
]
}
mutate {
convert => {
"SourceIP" => "string"
"DestinationIP" => "string"
"SourcePort" => "integer"
"DestinationPort" => "integer"
"Duration" => "float"
"FlowBytesSent" => "integer"
"FlowSentRate" => "float"
"FlowBytesReceived" => "integer"
"FlowReceivedRate" => "float"
"PacketLengthVariance" => "float"
"PacketLengthStandardDeviation" => "float"
"PacketLengthMean" => "float"
"PacketLengthMedian" => "float"
"PacketLengthMode" => "float"
"PacketLengthSkewFromMedian" => "float"
"PacketLengthSkewFromMode" => "float"
"PacketLengthCoefficientofVariation" => "float"
"PacketTimeVariance" => "float"
"PacketTimeStandardDeviation" => "float"
"PacketTimeMean" => "float"
"PacketTimeMedian" => "float"
"PacketTimeMode" => "float"
"PacketTimeSkewFromMedian" => "float"
"PacketTimeSkewFromMode" => "float"
"PacketTimeCoefficientofVariation" => "float"
"ResponseTimeTimeVariance" => "float"
"ResponseTimeTimeStandardDeviation" => "float"
"ResponseTimeTimeMean" => "float"
"ResponseTimeTimeMedian" => "float"
"ResponseTimeTimeMode" => "float"
"ResponseTimeTimeSkewFromMedian" => "float"
"ResponseTimeTimeSkewFromMode" => "float"
"ResponseTimeTimeCoefficientofVariation" => "float"
"DoH" => "boolean"
}
}
-- INSERT --
87,1
72%
Haleemah Anwary
File
Edit
View
Haleemah Anwary
hanwary1@myseneca.ca
SRT411NAA
111704201
Ln 4, Col 10
57 characters
100%
Windows (CRL
UTF-8

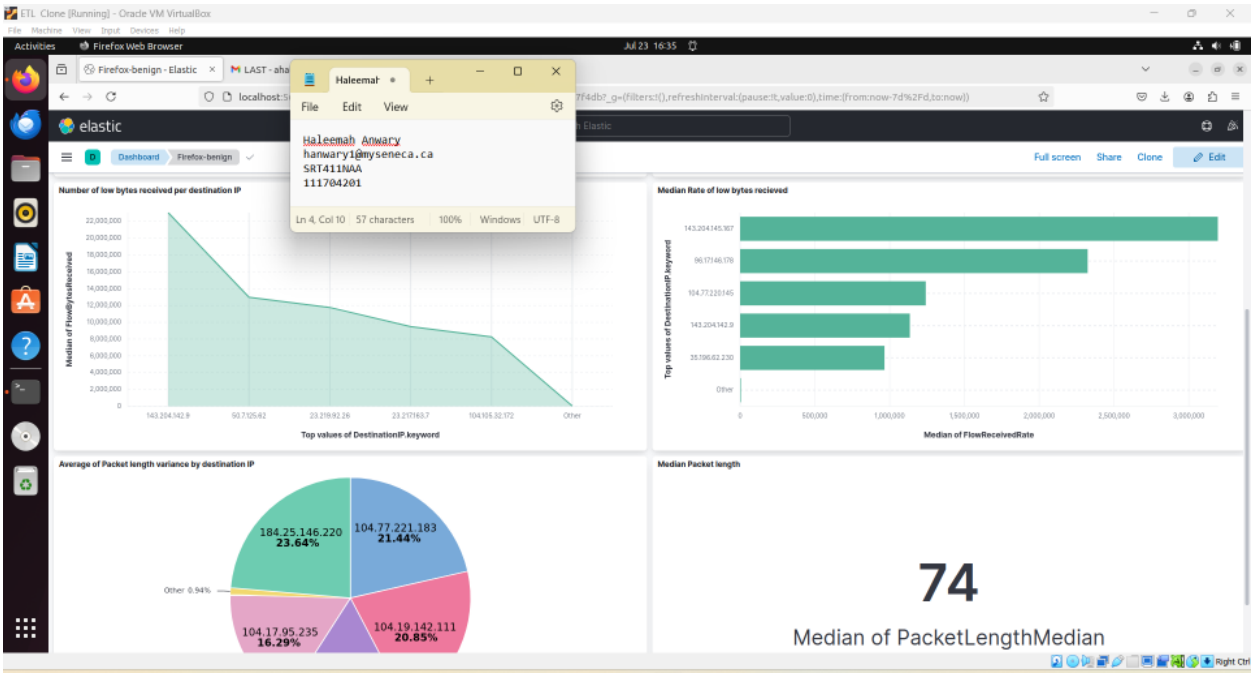

ETL Clone [Running] - Oracle VM VirtualBox
Activities
Terminal
Jul 23 16:10
ahalee@ETL: /etc/logstash/conf.d
"FlowSentRate" => "float"
"FlowBytesReceived" => "integer"
"FlowReceivedRate" => "float"
"PacketLengthVariance" => "float"
"PacketLengthStandardDeviation" => "float"
"PacketLengthMean" => "float"
"PacketLengthMedian" => "float"
"PacketLengthMode" => "float"
"PacketLengthSkewFromMedian" => "float"
"PacketLengthSkewFromMode" => "float"
"PacketLengthCoefficientofVariation" => "float"
"PacketTimeVariance" => "float"
"PacketTimeStandardDeviation" => "float"
"PacketTimeMean" => "float"
"PacketTimeMedian" => "float"
"PacketTimeMode" => "float"
"PacketTimeSkewFromMedian" => "float"
"PacketTimeSkewFromMode" => "float"
"PacketTimeCoefficientofVariation" => "float"
"ResponseTimeTimeVariance" => "float"
"ResponseTimeTimeStandardDeviation" => "float"
"ResponseTimeTimeMean" => "float"
"ResponseTimeTimeMedian" => "float"
"ResponseTimeTimeMode" => "float"
"ResponseTimeTimeSkewFromMedian" => "float"
"ResponseTimeTimeSkewFromMode" => "float"
"ResponseTimeTimeCoefficientofVariation" => "float"
"DoH" => "boolean"
}
}
date {
match => ["TimeStamp", "ISO8601"]
target => "@timestamp"
remove_field => ["TimeStamp"]
}
}
output {
elasticsearch {
hosts => ["http://localhost:9200"]
index => "firefox-benign"
}
stdout { codec => rubydebug }
}
-- INSERT --
109,1
Bot
Haleemah Anwary
File
Edit
View
Haleemah Anwary
hanwary1@myseneca.ca
SRT411NAA
111704201
Ln 4, Col 10
57 characters
100%
Windows (CRL
UTF-8

Data in Discover



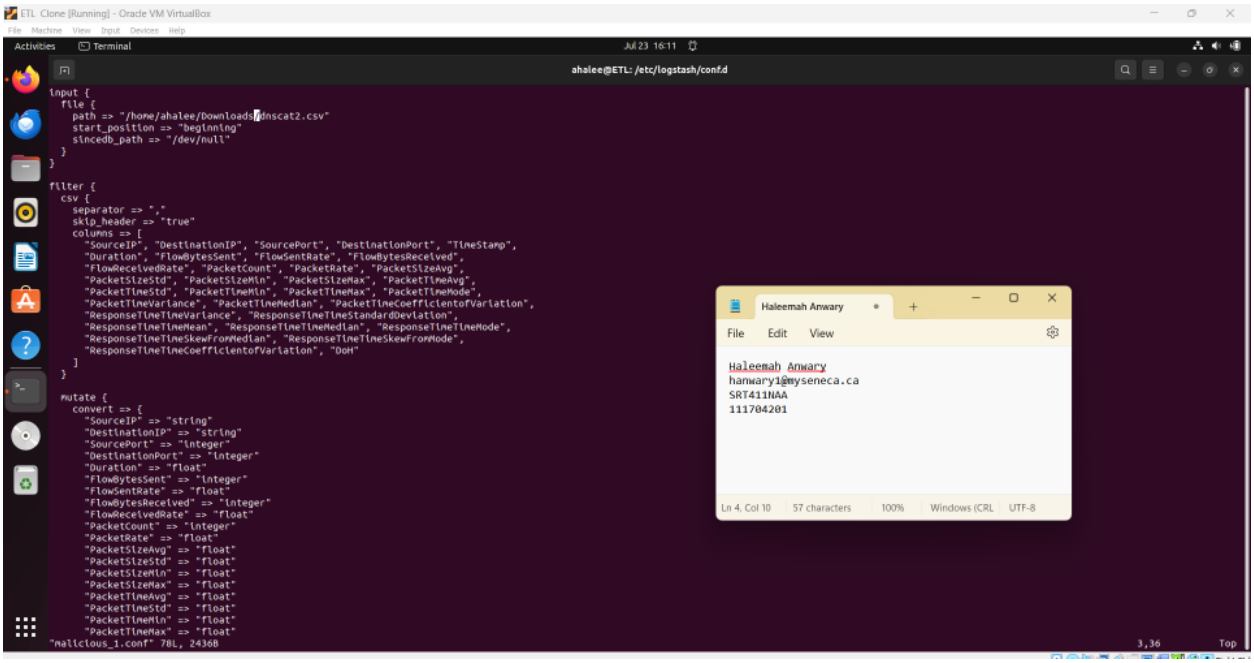
Visualization

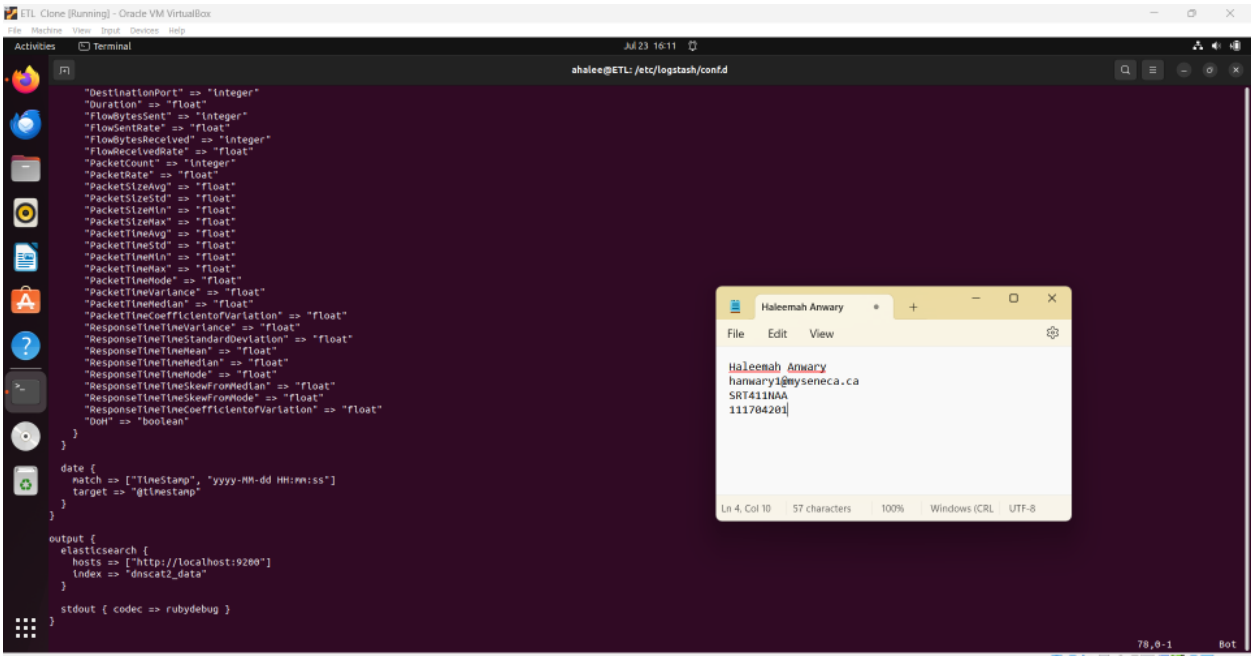




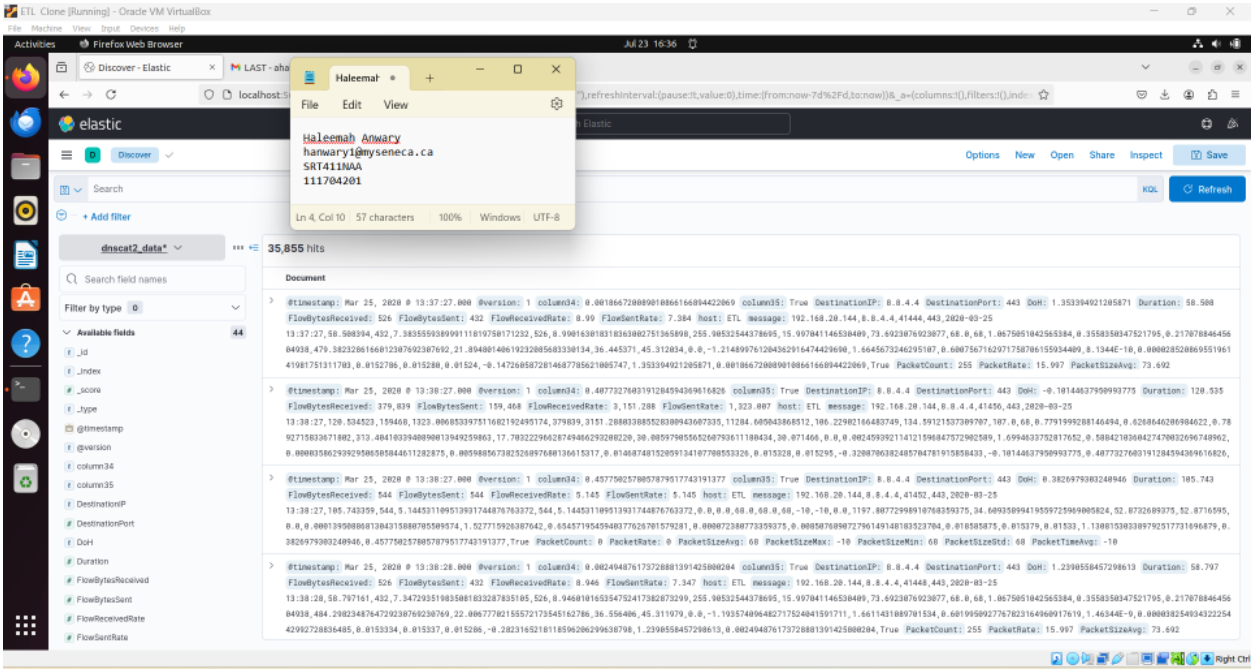
dnscat2 (malicious)

Logstash Pipeline

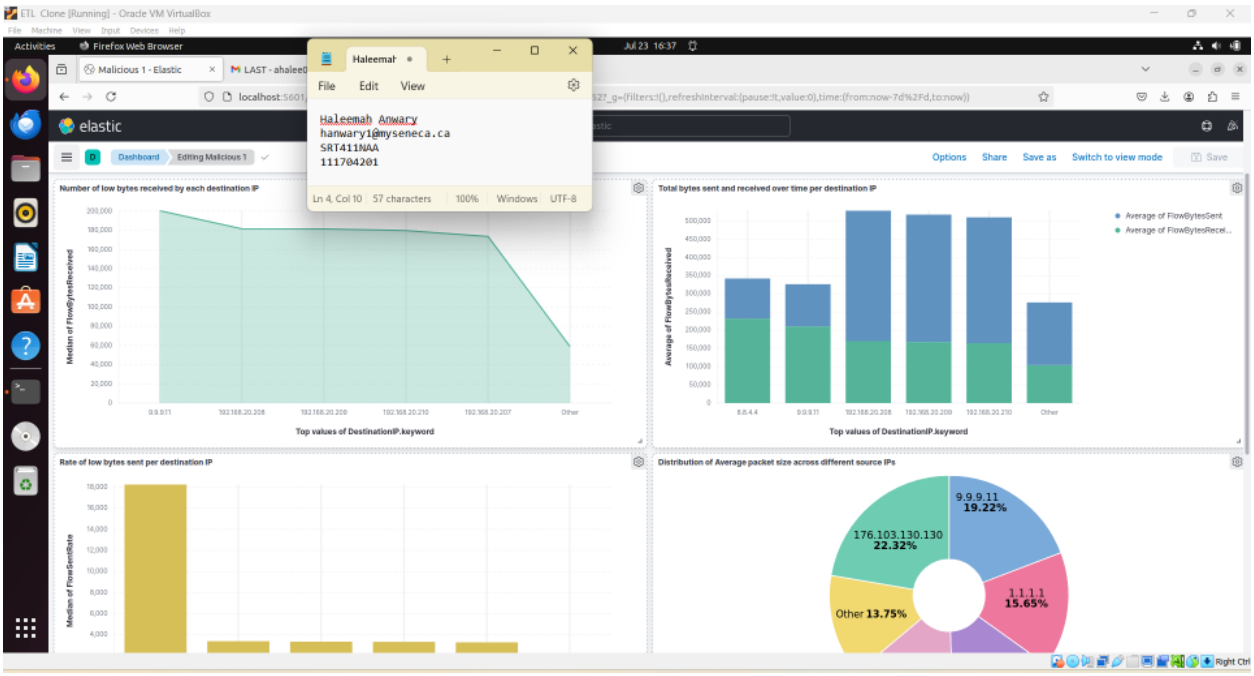


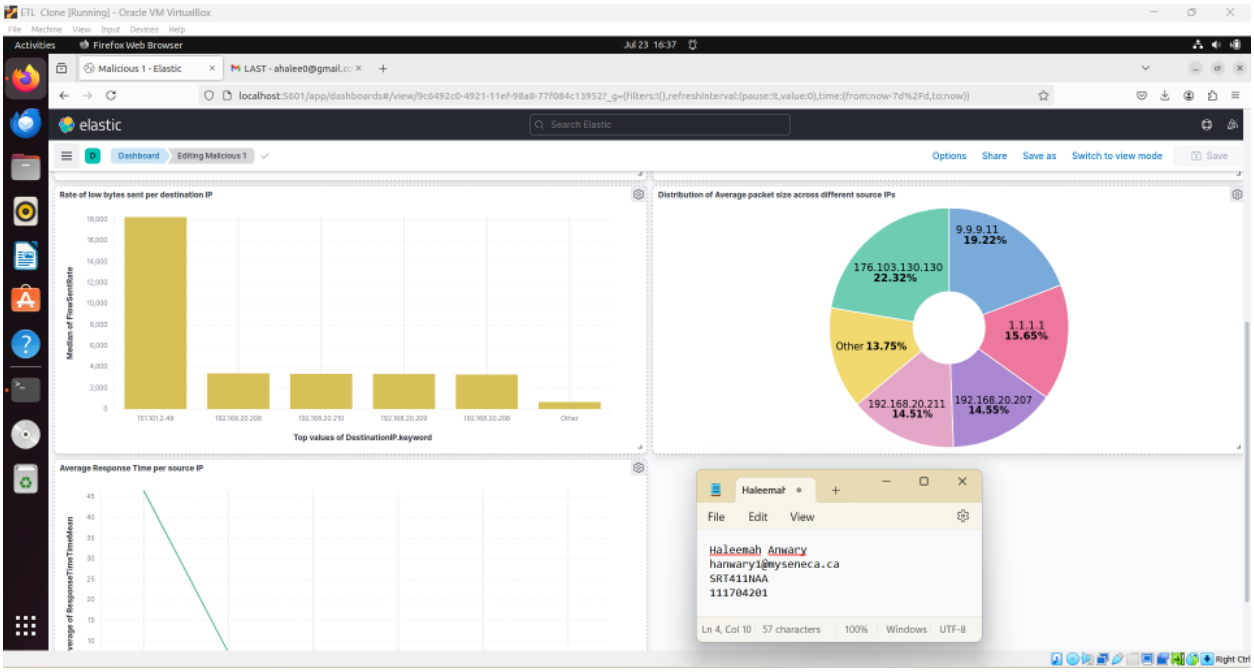


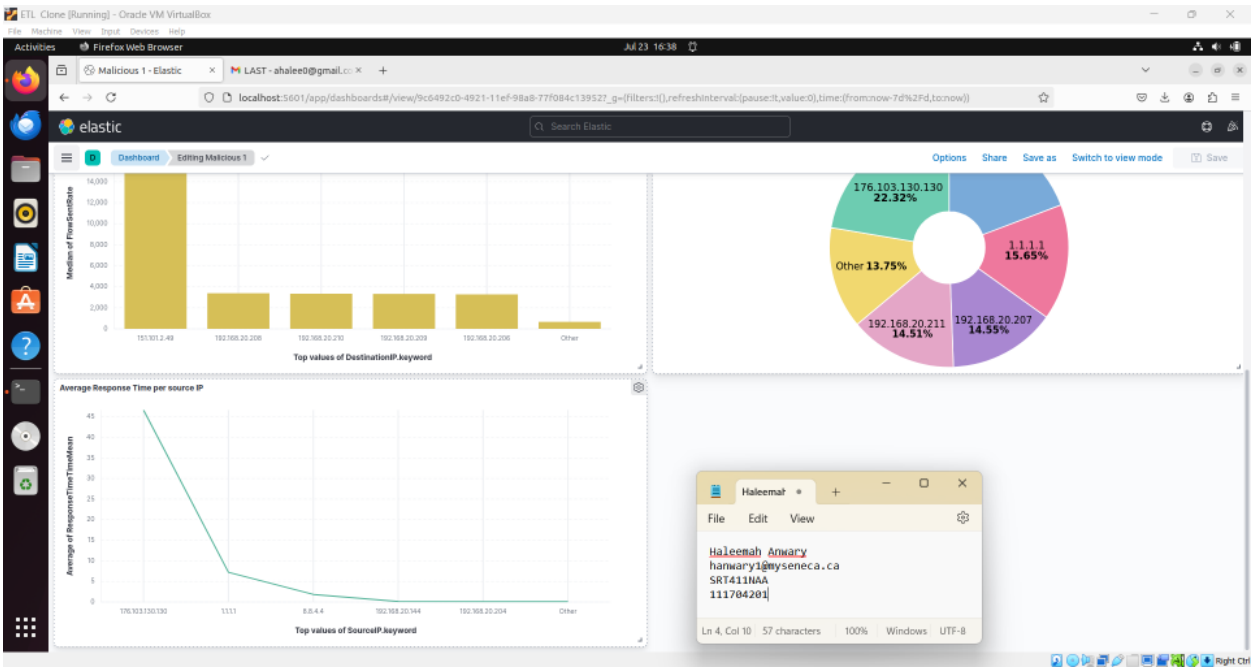
Data in Discover



Visualization

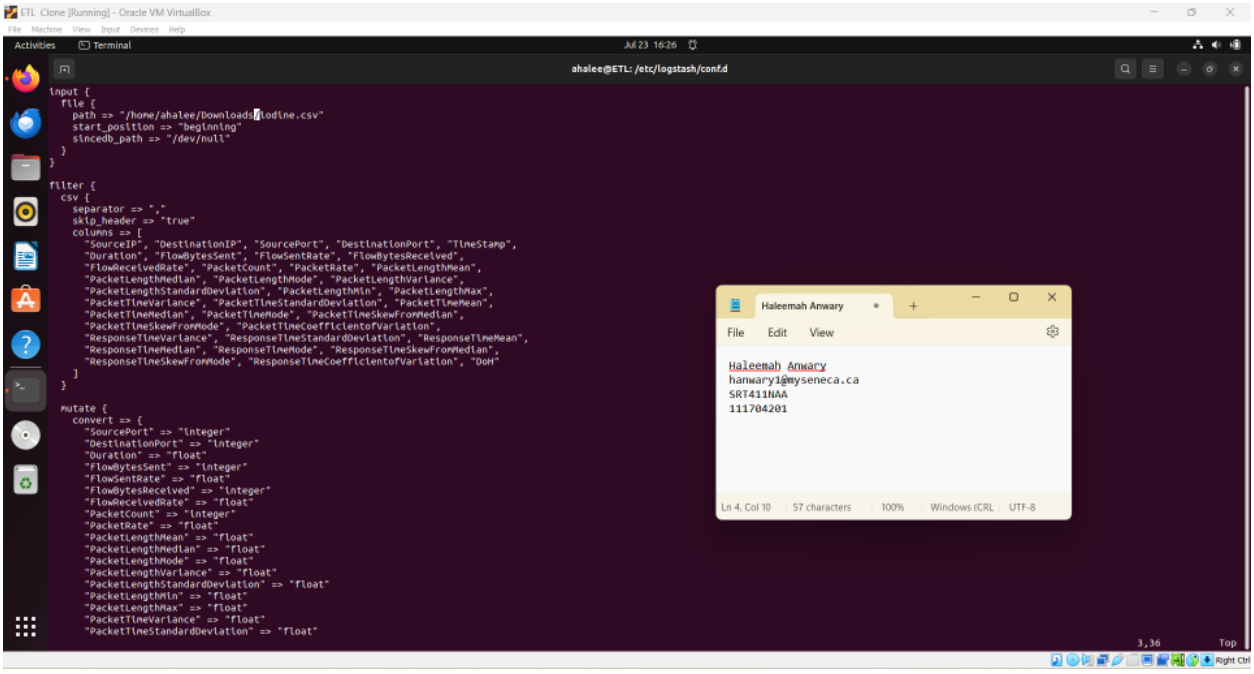


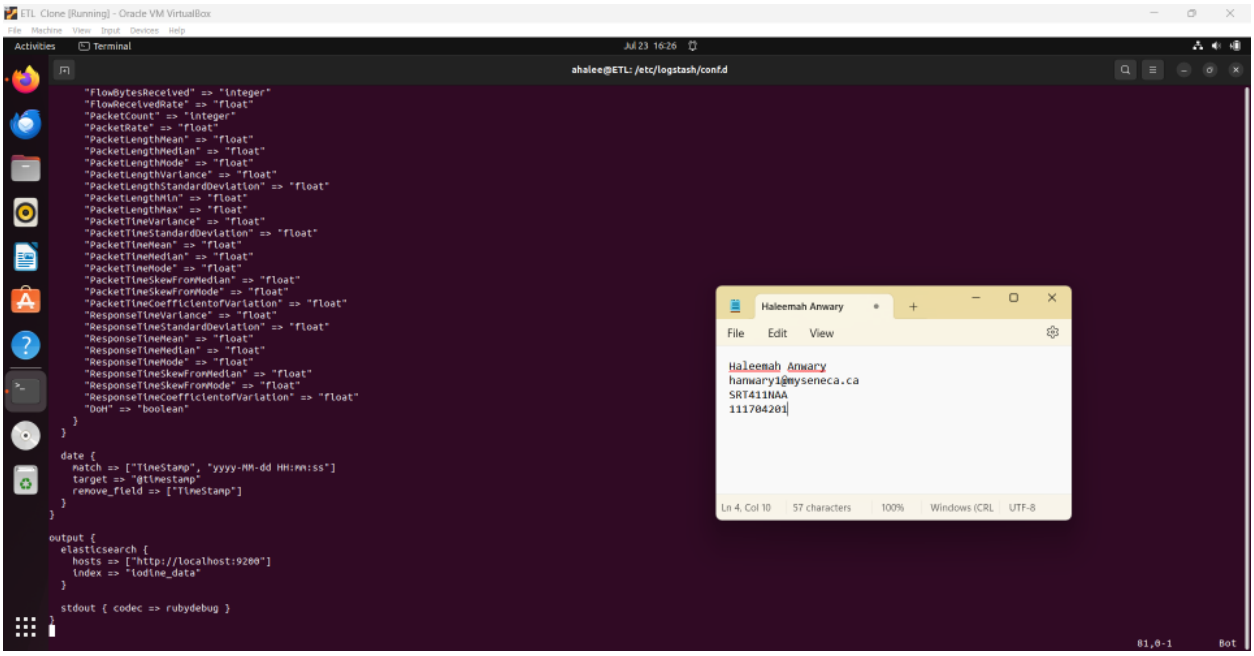




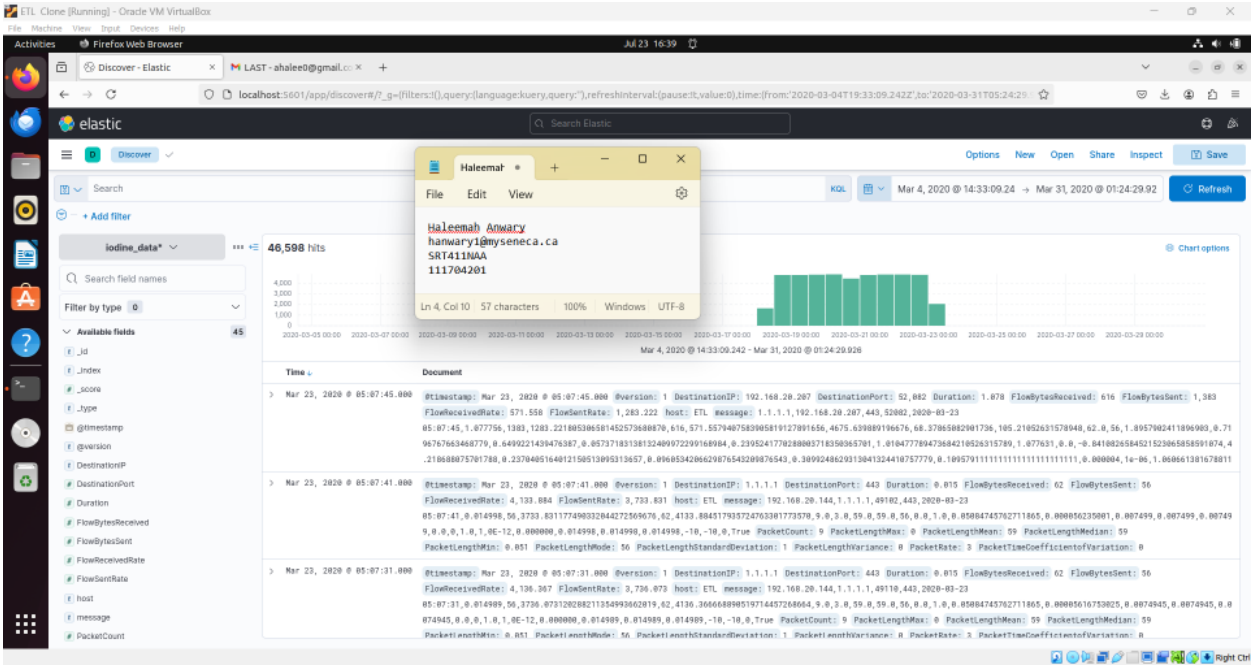
iodine (malicious)

Logstash Pipeline

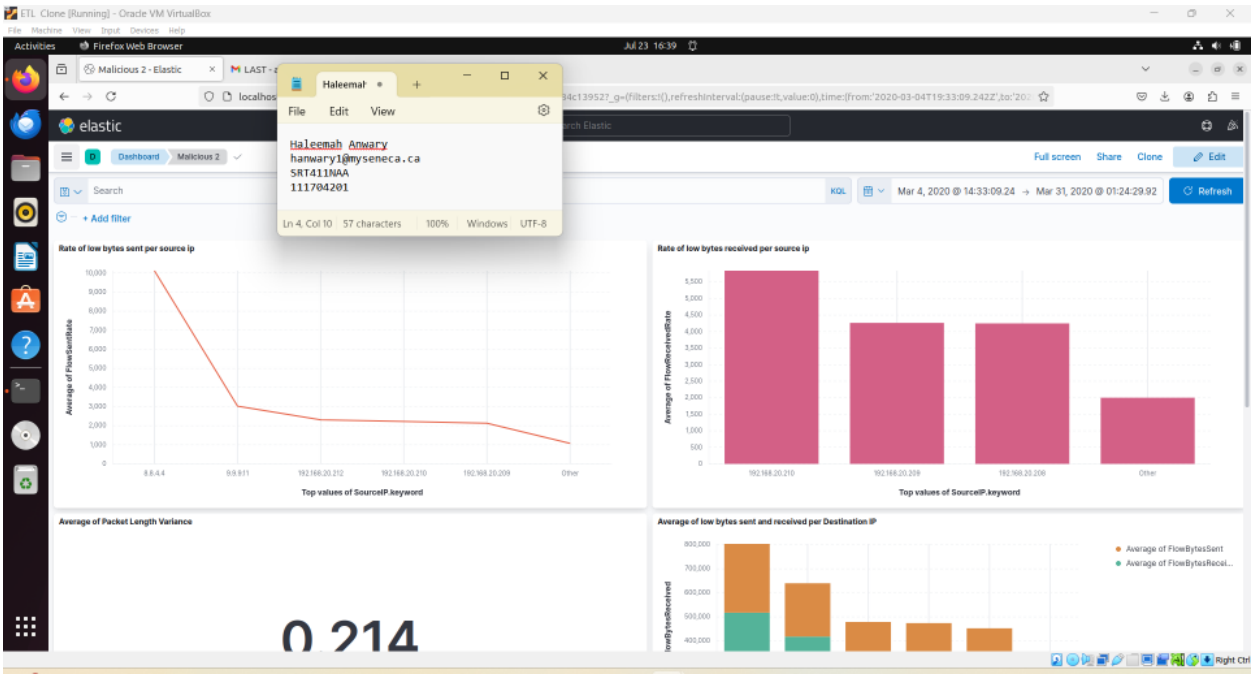


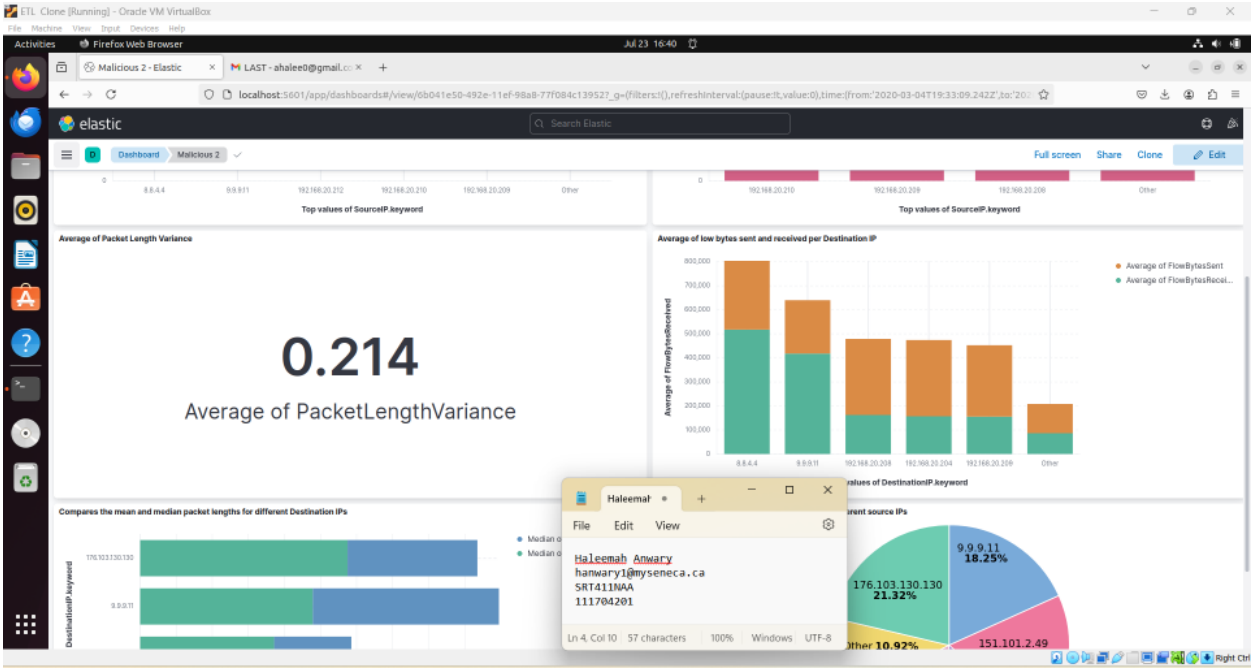


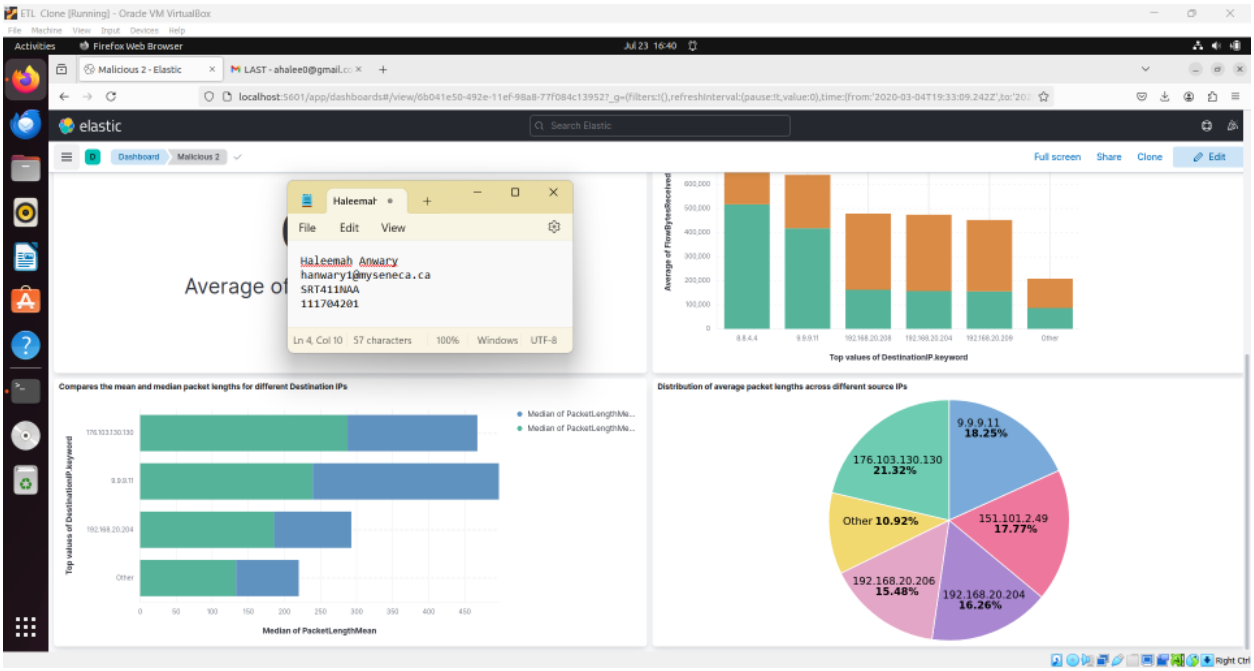
Data in Discover



Visualization






ETL Clone [Running] - Oracle VM VirtualBox
Activities
Firefox Web Browser
Jul 23 16:40
Malicious 2 - Elastic
LAST - ahalee0@gmail.com
localhost:5601/app/dashboards#/view/6b041e50-492e-11ef-98a8-77f084c139527_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:'2020-03-04T19:33:09.242Z',to:'20
elastic
Search Elastic
Dashboard
Malicious 2
Full screen
Share
Clone
Edit
Haleemah
File
Edit
View
Haleemah Anwary
hanwary1@myseneca.ca
SRT411NAA
111704201
Ln 4, Col 10
57 characters
100%
Windows
UTF-8
Average of
Average of FlowBytesReceived
Top values of DestinationIP.keyword
Compares the mean and median packet lengths for different Destination IPs
Median of PacketLengthMe...
Median of PacketLengthMe...
Top values of DestinationIP.keyword
Other
0
50
100
150
200
250
300
350
400
450
Median of PacketLengthMean
Distribution of average packet lengths across different source IPs
9.9.9.11
18.25%
176.103.130.130
21.32%
151.101.2.49
17.77%
Other 10.92%
192.168.20.206
15.48%
192.168.20.204
16.26%
Right Ctrl